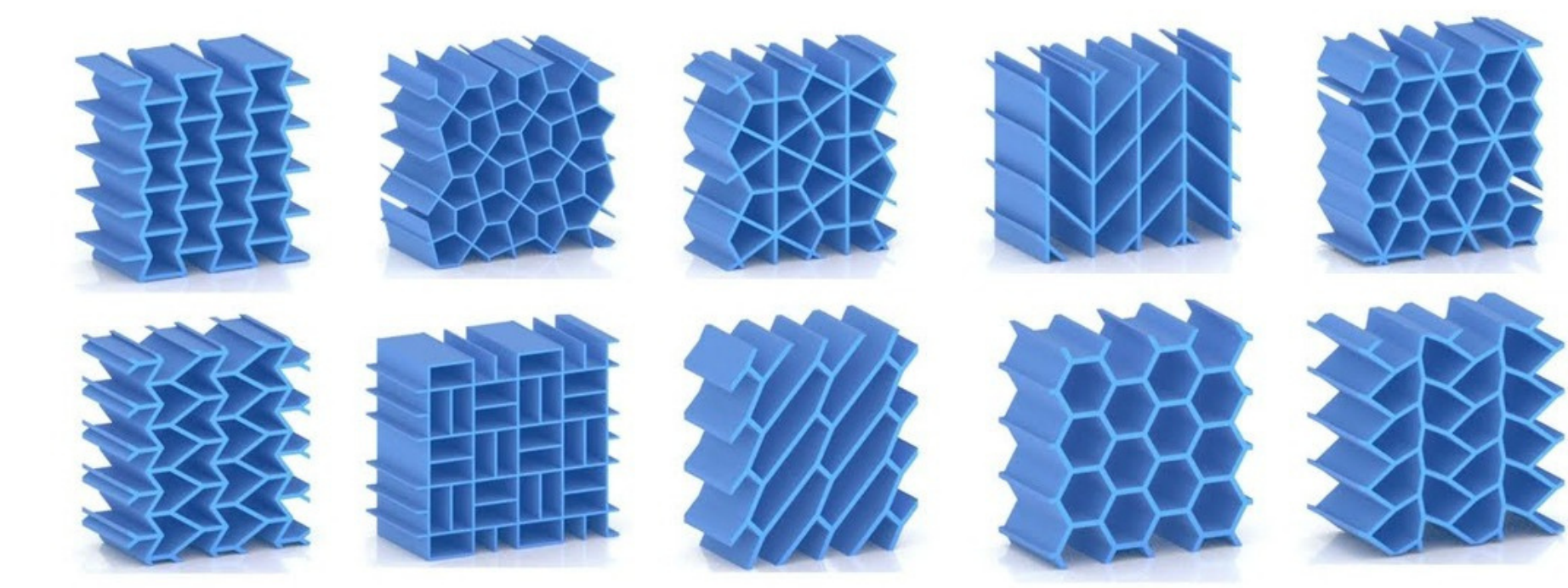
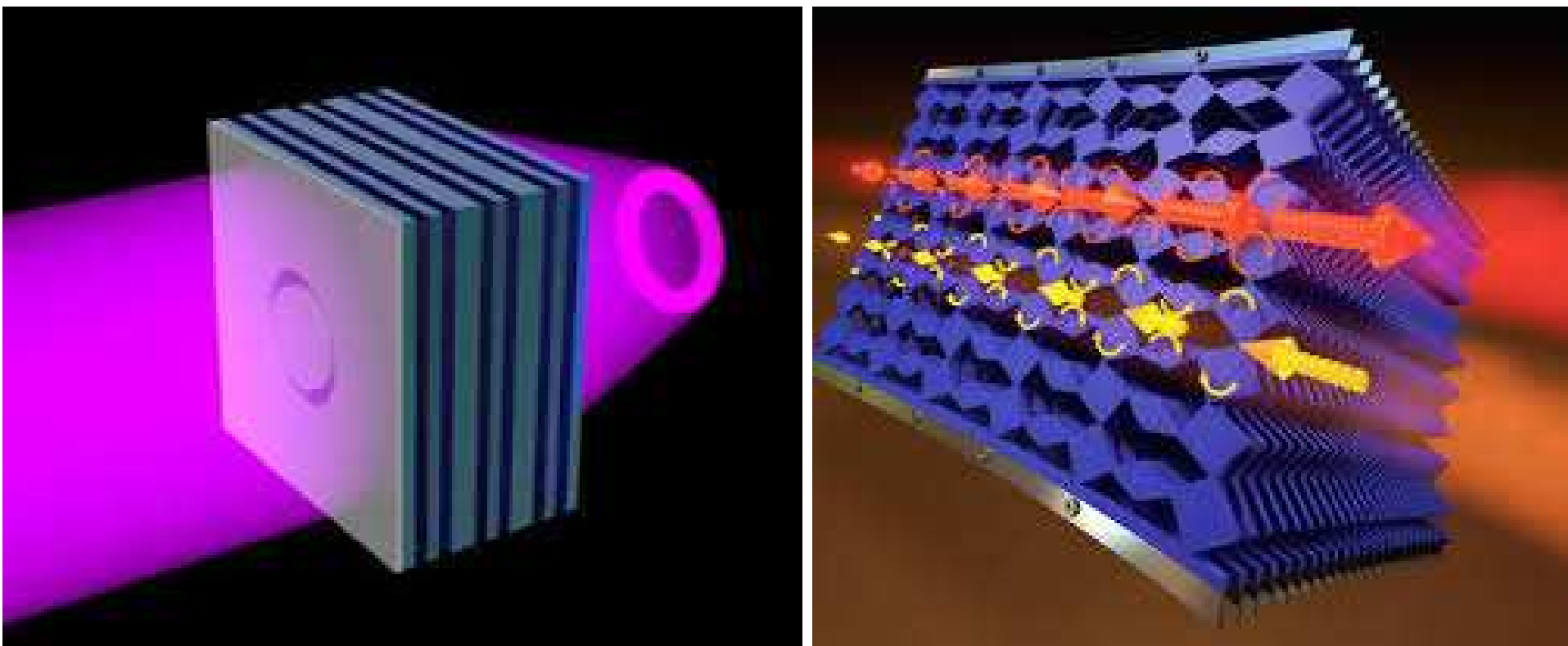




Introduction

- Metamaterials are artificially structured composites with subwavelength unit cells, enabling electromagnetic behaviors not found in nature (Smith et al., 2004; Jing et al., 2025).
- Their optical properties are dictated by the geometry, arrangement, and nanoscale coupling of their structures, rather than intrinsic material properties (Cai & Shalaev, 2010).
- Standard numerical methods like the Finite Element Method (FEM) are computationally expensive, which limits design space exploration and scalability (Hecht, 2012).
- A new machine learning pipeline that combines FEM datasets with generative models and surrogate learning to efficiently map the relationship between geometry and optical response.



Methods

- Developed a computational pipeline for inverse design of photonic metamaterials
- Generated parametric unit-cell geometries using three normalized variables controlling pore size, shape, and aspect ratio
- Meshed and evaluated each geometry using finite element methods (FEM) to compute energy loss across selected frequencies
- Trained a variational autoencoder (VAE) on the dataset of geometry-response pairs to compress high-dimensional geometry into a low-dimensional latent space
- Performed inverse design by optimizing within the latent space to match a target response, followed by decoding back to physical geometry
- Verified selected designs through additional FEM simulations

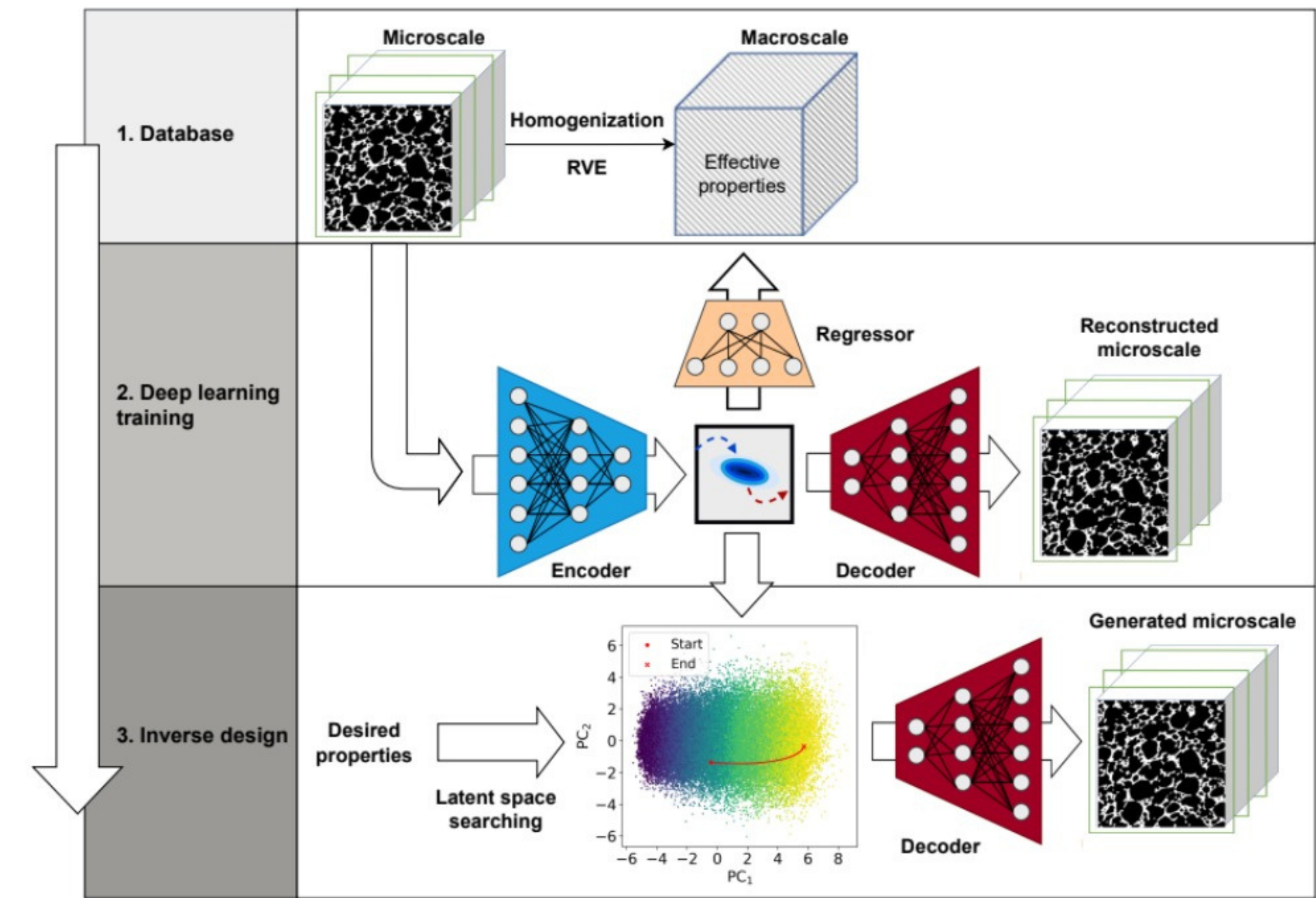
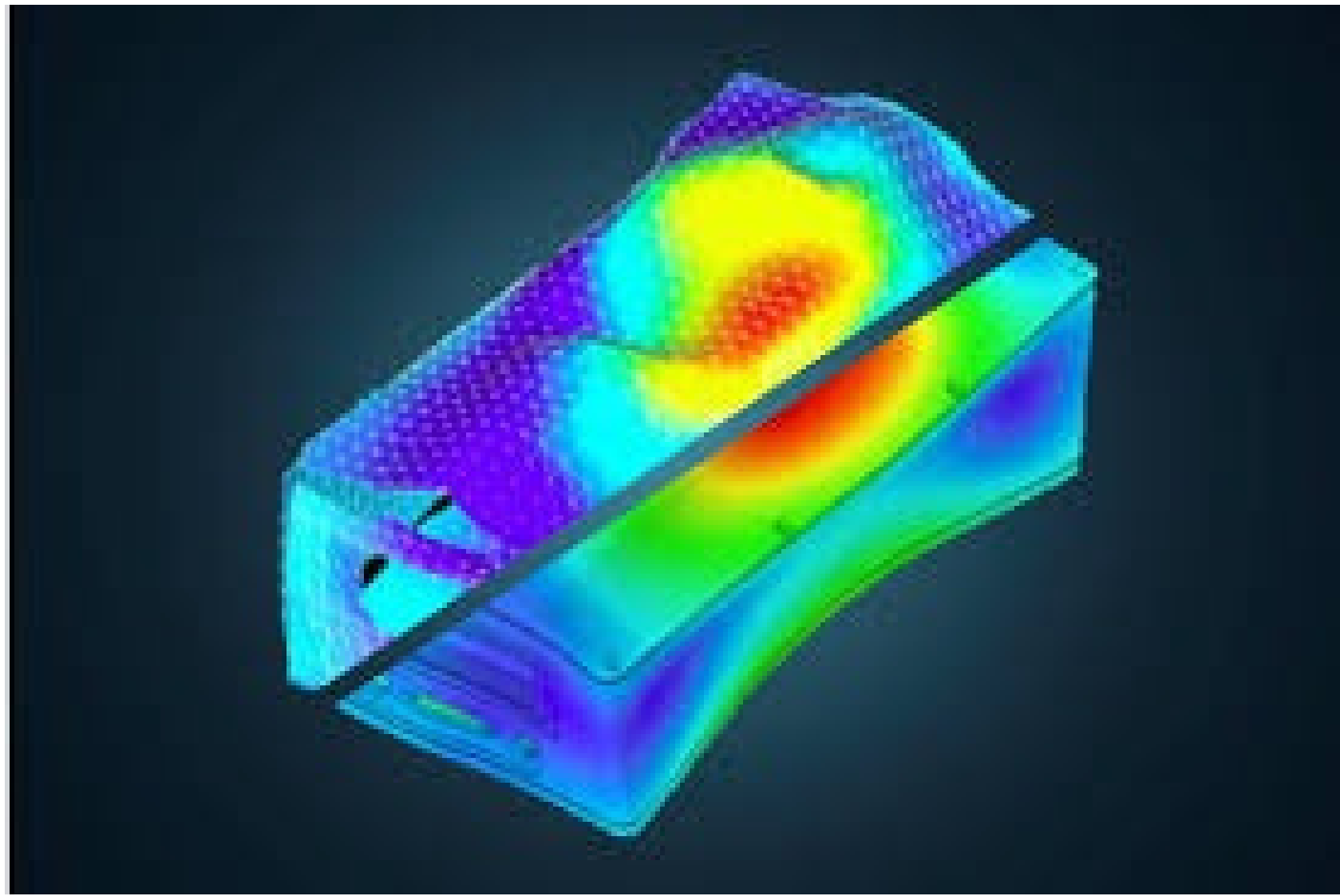


Figure 1. Machine Learning Pipeline Diagram for Inverse Design
Nguyen, P. T., Heider, Y., Kochmann, D. M., & Aldakheel, F. (2025)

Machine-Learning-Driven Inverse Design of Photonic Metamaterials Using FEM-Generated Spectral Data



Results

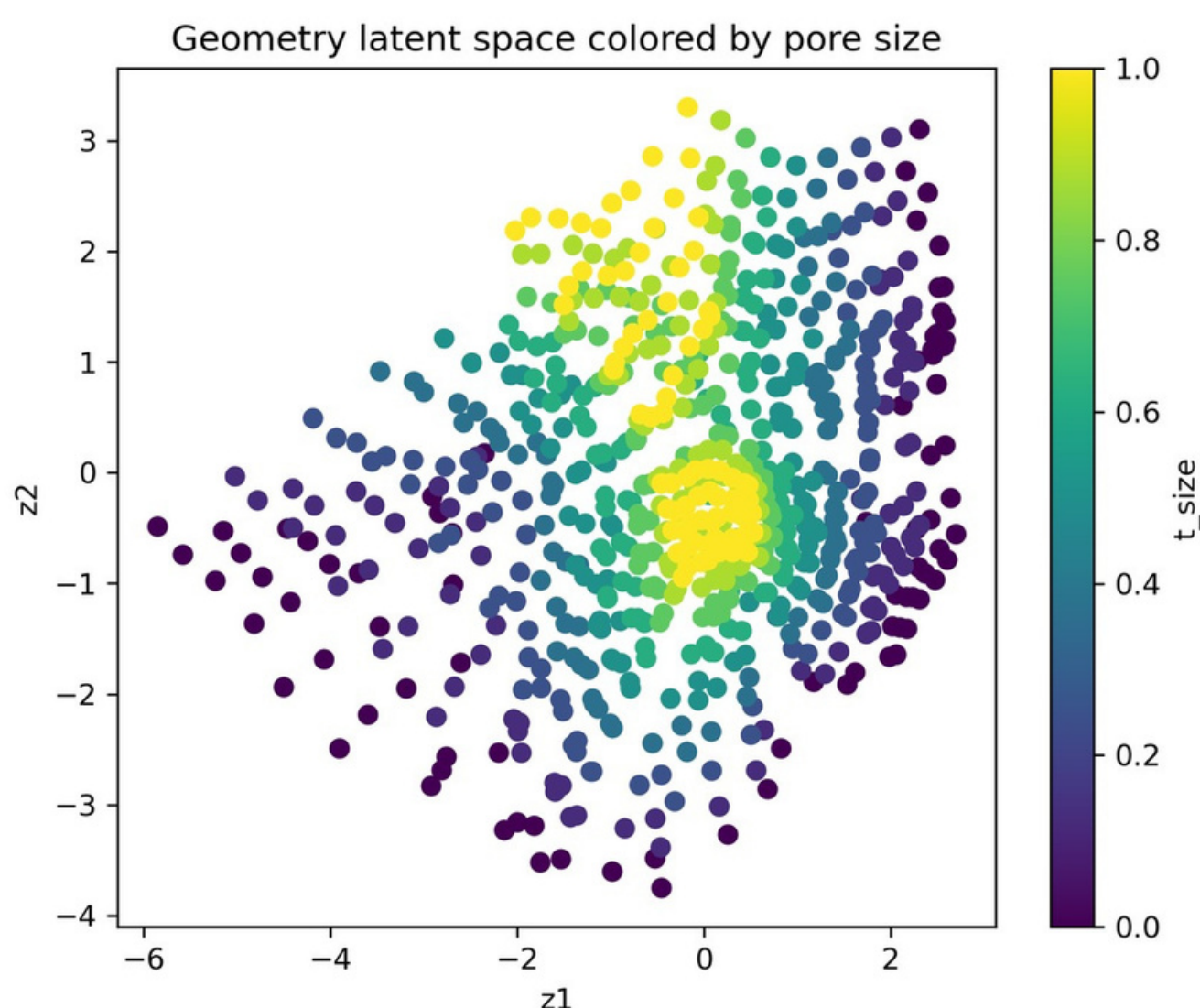


Figure 2. Latent Space Representation for Parametric Variables
Graph made by researcher through Google Colab

Metrics of Model Accuracy		
Object	R ²	RMSE
VAE (MLP)	0.9137	9.27E-02
Surrogate (MLP)	0.9035	1.01E-14
Verified Energy Loss	0.9983	5.66E-16

Table 1. Performance Metrics
Table made by researcher through google sheets

Graphs only show t_size for simplicity (3 variables total)

Energy loss of metamaterials is independent from frequency

Verified Loss Energy is from comparing FEM simulated values of decoded vs. training

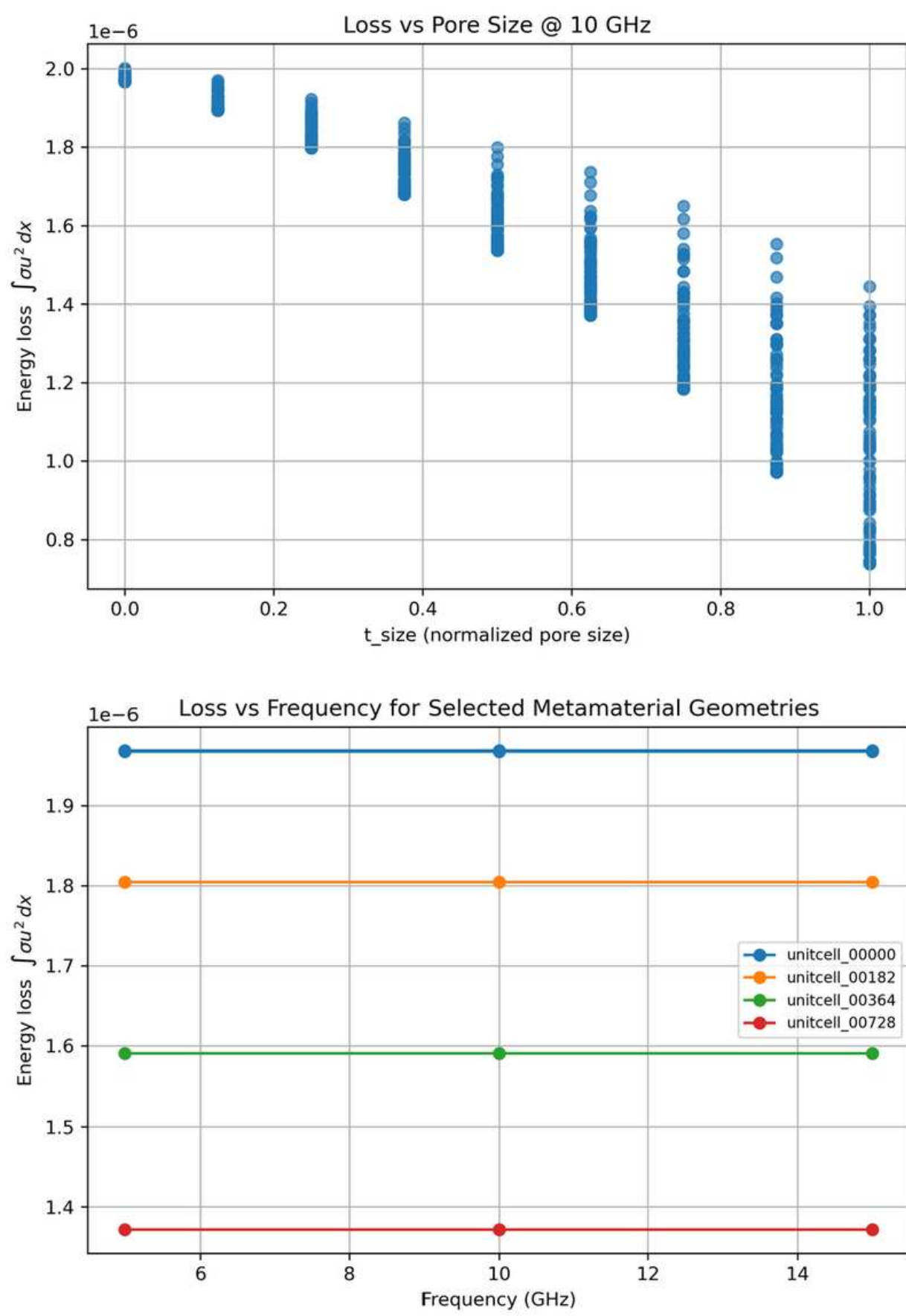


Figure 3. Comparison of Parametric Variables as well as Output vs Frequency
Graphs made by researcher through Google Colab

Discussion

- The proposed pipeline effectively captures the relationship between metamaterial geometry and electromagnetic response
- The VAE learns a structured latent space where geometries are smoothly organized according to physical parameters like pore size
- This enables efficient navigation of the design space and supports gradient-based optimization for inverse design
- The surrogate model enables rapid prediction of spectral responses, reducing reliance on expensive FEM simulations
- High agreement between surrogate predictions and FEM verification indicates the learned mapping is accurate and physically consistent
- Computed energy loss remains nearly independent of frequency, suggesting the system operates in a quasi-static regime
- This simplifies the learning problem and improves model stability but limits the ability to represent realistic optical behaviors
- Frequency-dependent phenomena such as resonance, interference, and dispersion are not captured
- This limits the framework's ability to represent advanced photonic applications

Limitations

Part 1: Scalar Approximation

- Scalar approximation of Maxwell's equations limits the model's ability to capture true electromagnetic wave behavior
- Results in frequency-independent responses, reducing physical realism and applicability to real-world optical systems

Part 2: Low-dimensional Geometry

- Low-dimensional geometry representation (3 parameters) constrains diversity and complexity of achievable designs
- More expressive representations needed for practical applications

Part 3: Dataset and Optimization

- Simulation-based dataset risks overfitting and limits generalization to fabricated structures
- Mesh generation inconsistencies cause invalid samples, highlighting robustness issues
- Single-objective optimization (energy loss) may not capture full range of desired optical properties in real applications

Future Research

- Future work will focus on improving the physical realism and modeling capability of the pipeline. A key direction is incorporating full Maxwell-based electromagnetic simulations to capture frequency-dependent behaviors such as resonance and wave propagation.
- Additionally, more expressive geometry representations (e.g., image- or voxel-based encodings) will enable more complex and realistic designs. Physics-informed learning methods can further improve model consistency by embedding physical constraints into training.
- Finally, expanding the dataset and extending the framework to multi-objective optimization will allow simultaneous control of multiple optical properties, improving applicability to real-world metamaterial design.

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